

1. Project Overview

Problem Statement: Malaysia's government digital services were built to reduce bureaucracy, but for millions of citizens, the friction has simply moved online. Users are forced to navigate fragmented, complex portals (MyTax, PTPTN, MySikap), resulting in repetitive data entry and confusing navigation. Furthermore, uploading unprotected personal documents (like an NRIC) creates severe anxiety regarding identity theft and data misuse. For seniors, first-time taxpayers, and busy professionals, these disjointed systems act as barriers rather than conveniences—creating a distinct "Digital Bureaucracy Gap."

Target Domain: Streamlining digital e-government services, automating complex administrative workflows, and enforcing client-side document security.

Proposed Solution Summary: GovEase AI is a multi-agent ecosystem that acts as a personal digital liaison, removing the friction of bureaucratic portals. Powered by the Z AI Model, it enables users to submit their details via a simple, familiar chat interface. The system automatically extracts data, securely watermarks sensitive documents locally, and autonomously fills out lengthy forms across various government websites. This eliminates repetitive data entry and shifts the citizen's role from an anxious manual form-filler to an empowered final reviewer.

2. Background & Business Objective

Background of the Problem: For years, Malaysia's digital government transformation focused on building standalone portals for individual agencies (LHDN, PTPTN, KWSP, JPJ). This created a fragmented, siloed ecosystem. Citizens are currently forced through a highly repetitive, manual pathway: logging into multiple disparate websites, deciphering complex bureaucratic jargon, manually typing the exact same demographic data into different forms, and uploading unprotected, highly sensitive documents like their National Registration Identity Card (NRIC). This persistent digital friction—the "Digital Bureaucracy Gap"—disproportionately affects senior citizens, busy professionals, and first-time users, effectively gatekeeping essential services behind a wall of technical and administrative fatigue.

Importance of Solving This Issue: Solving this gap transforms the fundamental relationship between the citizen and the state. By compressing a multi-portal, multi-step data entry process into a single, intuitive chat conversation, GovEase AI reduces transaction times from hours to mere seconds. Furthermore, by introducing automated client-side document watermarking (The Guardian), we address a critical, widespread anxiety regarding identity theft and data privacy. Ultimately, solving this issue ensures that technical literacy or patience for form-filling is no longer a prerequisite for accessing government services—allowing citizens to focus purely on review and consent.

Strategic Fit / Impact: GovEase AI aligns perfectly with Malaysia's broader GovTech digital transformation goals, such as the MyDIGITAL blueprint and the MADANI framework, which advocate for accessible, citizen-centric public services. By placing the **Z AI Model** at the core of a multi-agent ecosystem, it demonstrates how advanced reasoning models can bridge the gap between rigid legacy infrastructure and modern user expectations. The impact is twofold: it drastically reduces the administrative burden and privacy anxieties for millions of citizens, while potentially increasing on-time compliance for government agencies (like LHDN and PTPTN) by removing the friction that leads to user drop-off.

3. Product Purpose

3.1. Main Goal of the System: To provide a secure, AI-powered personal digital liaison that eliminates the friction of government bureaucracy by automating complex data entry and protecting citizen privacy.

3.2. Intended Users (Target Audience):

- Product Manager
- Founder of Products (Early Stage)
- Entrepreneurs
- UI/UX Designer
- Frontend Developers
- Low Code Builders

4. System Functionalities

4.1. Description: The system operates as a multi-agent autonomous workflow designed to streamline complex government interactions. Powered by the Z AI Model, the system understands natural language and unstructured documents, securely processes personal data locally, and intelligently maps the extracted information directly into the rigid web forms of various government portals.

4.2. Key Functionalities:

- **Conversational Data Extraction (The Concierge):** Translates natural language chat and document photos (e.g., IC, EA forms) into structured data using high-accuracy OCR, securely storing the extracted fields directly on the user's local device.
- **Dynamic Document Watermarking (The Guardian):** Automatically applies agency-specific, indelible watermarks (e.g., "FOR LHDN USE ONLY") to personal documents locally before any internet transmission, ensuring absolute data security and preventing misuse.
- **Autonomous Form Autofill (The Executor):** Acts as a browser co-pilot that detects the active government portal, intelligently maps the user's stored data to the correct input fields, and populates the entire application in seconds.
- **Human-in-the-Loop Review:** Automatically pauses the workflow at the end of data entry, allowing the user to review the auto-filled information and manually click the final submit button, ensuring the user retains full control and legal responsibility.

4.3. AI Model & Prompt Design

4.3.1. Model Selection: The core engine powering GovEase AI is the **Z AI GLM**. This model was selected because its reasoning capabilities are highly optimized for three critical tasks required by our product:

1. **Bilingual Semantic Parsing:** It accurately understands code-switching between English and Bahasa Melayu, which is essential for parsing Malaysian user inputs and government portal jargon.
2. **Vision-Language Processing:** It excels at reading unstructured text from uploaded documents (OCR) such as NRICs or EA forms.
3. **Structured Data Output:** It reliably generates strict JSON schemas required by the Executor agent to map data into HTML form fields without syntax errors.

4.3.2. Prompting Strategy: We utilize **Multi-step Agentic Prompting** combined with **Few-Shot Learning** for data extraction.

- *Reasoning:* A single prompt asking the model to "chat with the user, read their IC, and fill the form" is prone to hallucination. Instead, we break the process down. The Concierge uses zero-shot prompting for conversational routing ("What does the user want to do?"). Once an intent is identified, a specialized extraction prompt is triggered using few-shot examples of Malaysian documents to ensure high-accuracy data structuring. This multi-step strategy isolates tasks, significantly reducing the cognitive load on the model and preventing data cross-contamination.

4.3.3. Context & Input Handling: The system primarily handles unstructured conversational text and image uploads.

- *Size Limits & Compression:* The maximum input size for document uploads is capped at 5MB per image. If an image exceeds this size, the client-side frontend automatically compresses it before transmission. For text, the conversational context window is capped at 4,000 tokens per session.
- *Truncation & Rejection:* If the text context exceeds the limit, the system employs a "sliding window" approach—older conversational turns are truncated, but the core extracted JSON state is preserved. If an uploaded file is completely unreadable or in an unsupported format, the system rejects the input instantly, skipping the model call to save compute resources.

4.3.4. Fallback & Failure Behavior: Because GovEase AI handles sensitive bureaucratic data, we cannot afford hallucinations (e.g., inventing a fake IC number).

- *Schema Validation:* All model outputs are passed through a strict schema validator (e.g., Zod). If the model returns an unusable or malformed response (like an IC number without the standard YYMMDD-XX-XXXX format), the system automatically triggers a silent retry with a lower temperature setting for stricter adherence.
- *Graceful Error State:* If the retry fails (e.g., the photo is simply too blurry), the system falls back to a conversational error state. The Concierge will reply: *"I couldn't read the details on that document clearly. Could you please upload a clearer photo, or manually type your IC number?"*
- *Human-in-the-Loop:* Ultimately, the product's architecture relies on "Step 4: You Stay in Control." The AI is restricted from clicking the final "Submit" button on any portal. If an edge-case hallucination makes it to the browser, the user serves as the final escalation path during the manual review phase.

5. User Stories & Use Cases

User Stories:

- "As a first-time taxpayer, I want to upload a photo of my EA form into a familiar chat interface so that the system extracts my details without me having to manually decipher the complex MyTax portal."
- "As a security-conscious citizen, I want my uploaded IC to be automatically watermarked for the specific agency (e.g., LHDN) before it leaves my device so that my identity cannot be stolen or misused."
- "As a senior citizen, I want my personal details to automatically populate on government websites so that I only have to review the information and click 'Submit' without typing into tiny form fields."

Use Cases (Main Interactions):

- **Conversational Extraction Flow:** The user uploads a document (e.g., NRIC or salary slip) or types a natural language request into the chat. The Concierge (powered by the Z AI Model) parses the intent, performs OCR on the document, extracts the required structured fields, and saves the data securely to the local device.
- **Secure Watermarking Flow (The Guardian):** When a user uploads a sensitive document, The Guardian intercepts the image locally. It automatically applies an indelible, context-specific watermark (e.g., "FOR LHDN USE ONLY") directly onto the image via a client-side canvas before it is ever used for submission.
- **Portal Autofill & Review Flow (The Executor):** The user navigates to a supported government portal. The Executor browser extension detects the target URL, retrieves the locally stored JSON data, and instantly maps the values to the corresponding website DOM input fields. It then pauses, allowing the user to review the populated form and manually execute the final submission.

6. Features Included (Scope Definition)

- A conversational chat interface (The Concierge) allowing users to easily input data and upload documents using natural language.
- An automated client-side security layer (The Guardian) that dynamically applies context-specific watermarks to sensitive documents (e.g., NRIC) before any transmission.
- A browser extension co-pilot (The Executor) that intelligently detects active government portals and autonomously maps stored data into web forms.
- Support for both text-to-data semantic parsing and picture-to-data (OCR) extraction, powered entirely by the Z AI Model.
- A mandatory "Human-in-the-Loop" review checkpoint that pauses the automation, ensuring the user retains final control over the actual submission.

7. Features Not Included (Scope Control)

- **Direct Backend API Integration:** The system functions entirely via client-side DOM manipulation (The Executor) and local data processing. Direct integration with government backend databases or APIs is strictly out of scope due to security protocols and access limitations.
- **Production Environment & Legal Liability:** This MVP is a functional proof-of-concept for the hackathon. It is not intended for immediate production deployment, and it does not assume legal liability for the accuracy of tax filings or government submissions (hence the mandatory manual review step).
- **Multi-User & Corporate Filings:** The initial version focuses solely on single-user, individual citizen profiles. Multi-user concurrent inputs, corporate tax filings, or enterprise account management are not supported.
- **Payment Gateway Integration:** While the system will automate the form-filling process up to the final step, executing actual financial transactions (e.g., paying for the road tax or PTPTN loan) via integrated payment gateways is out of scope for this phase.

8. Assumptions & Constraints

LLM Cost Constraint: * **Estimation:** We estimate an average user session (one document upload + conversational parsing) will consume approximately 1,500 input tokens and 500 output tokens for the Z AI Model.

- **Cost Management Decision:** To keep inference costs manageable at scale, the system architecture specifically isolates the LLM usage to the "Concierge" step (upfront OCR and JSON extraction). The "Executor" (browser extension) relies on deterministic, zero-cost DOM mapping to fill the forms. We do not use the LLM to continuously read or navigate the browser state during form filling, effectively cutting out repetitive API calls.

Technical Constraints: * **DOM Dependency:** The Executor browser extension relies heavily on the structural consistency of government websites. If a portal (e.g., MyTax or PTPTN) undergoes a massive UI overhaul or obscures its input fields using complex Shadow DOMs or iFrames without accessibility labels, the autofill mapping may temporarily break.

- **Client-Side Limits:** The Guardian's watermarking tool is constrained to HTML5 Canvas capabilities. It provides strong visual watermarking but does not embed deep cryptographic steganography into the file metadata.

Performance Constraints: * **Image Processing Latency:** Processing high-resolution images for OCR via the Z AI Model can introduce latency. To maintain a responsive chat experience, client-side compression forces all uploaded documents to be under 5MB. If a user uploads a highly compressed or blurry image, OCR extraction accuracy will degrade, prompting a manual retry.

User Input: * **Mandatory Human Approval:** GovEase AI is an administrative assistant, not an autonomous legal entity. The system is strictly programmed to stop at the final review screen. It relies completely on the user to manually verify the accuracy of the auto-filled data and click the final "Submit" button.

- **Input Clarity:** The model cannot invent missing information. If a user uploads an incomplete document (e.g., a salary slip missing the employer's tax number), the system will halt and prompt the user to provide the missing data manually via chat.

9. Risks & Questions Throughout Development

- **Design Similarity:** [e.g. How are we ensuring that AI is not generating similar designs across users?]

To ensure GovEase AI avoids generating repetitive or identical outputs across different users, the system relies on Contextual Dynamic Injection. Since the Z AI Model processes unstructured data unique to each citizen—such as varying income levels, specific tax relief eligibility, or different employment backgrounds—the resulting JSON "action plan" is naturally distinct for every individual. Furthermore, the Executor agent performs real-time reasoning on the live DOM of the government portal, which varies based on the user's specific account status or outstanding balances. By combining these unique personal data points with a conversational interface that adapts its tone to the user's specific queries, the AI ensures that every interaction and form-filling sequence is a personalized response rather than a generic, one-size-fits-all template.

- **Frame Stability:** [e.g. Does the user need to fix the layout if it is found broken after exporting?]

The system maintains stability by using **non-destructive autofill**, meaning it populates text fields and checkboxes without refreshing the page or breaking the website's original layout. Because government portals can be visually inconsistent, the **Executor agent** performs a real-time "integrity check" after filling to ensure the data is placed in the correct boxes. If the mapping is found to be "broken" or visually misaligned due to a portal update, the user does not need technical skills to fix it; they simply use the **Human-in-the-Loop** review step to manually correct the text directly on the screen before clicking submit, ensuring the final submission is perfectly accurate regardless of the AI's initial placement.

- **Drawing Interpretation:** [e.g. What will happen if the AI is unable to contextualize the sketch that is provided?]

If the **Z AI Model** encounters an image it cannot interpret—such as a blurry NRIC, a coffee-stained salary slip, or ambiguous handwriting—it will not attempt to "guess" the data, which prevents hallucinations. Instead, the system triggers an immediate **Clarification Event** within the chat interface, highlighting the specific document or section it finds confusing. The user is then prompted to provide the missing context either by uploading a clearer photo or manually typing the information. By treating the AI as an assistant that seeks confirmation rather than a black box that operates in isolation, the system ensures that document interpretation remains grounded in reality and user-verified truth.

